



Cloud Computing

Assignment-Week 3

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total mark: 10 X 1 = 10

QUESTION 1:

Which of the following is least likely to be specified in an operational cloud Service Level Agreement(SLA)?

- A. Recovery Time Objective (RTO)
- B. Exact encryption algorithm and key storage procedures
- C. Mean Time To Repair (MTTR) for incidents
- D. Penalty / credit calculation for downtime

Answer: B

Detailed Solution : SLAs describe service-level commitments (RTO, MTTR, penalties); low-level implementation/security procedures (exact algorithms/key storage) belong in security/operational documents, not the SLA.

QUESTION 2:

A multilevel SLA generally consists of which of the following levels?

- (i) Corporate Level
- (ii) Customer Level
- (iii) Service Level

- A. Only (i)
- B. (i) and (ii)
- C. (ii) and (iii)
- D. (i), (ii), and (iii)

Correct Answer: D

Detailed Solution: A multilevel SLA typically consists of corporate-level commitments applicable to all customers, customer-level agreements tailored to specific clients, and service-level agreements defining guarantees for individual services.

QUESTION 3:

An enterprise signs a three-tier multilevel SLA with a cloud provider. However, the corporate-level



SLA guarantees 95% uptime, the customer-level SLA specifies 98% uptime, and the service-level SLA guarantees 99.9% uptime. In case of a service disruption, which level of SLA is the provider LEGALLY BOUND to honor?

- A. Corporate-level SLA, as it's the broadest agreement
- B. Customer-level SLA, as it's the middle ground
- C. Service-level SLA, as it's the most specific agreement
- D. The provider can choose the lowest applicable SLA

Correct Answer: C

Detailed Solution : Multilevel SLAs are structured hierarchically. The service-level SLA is the most granular and specific to the actual service delivery and is legally binding. If there's a conflict, the highest level of commitment (service-level SLA at 99.9%) applies. Corporate and customer-level SLAs provide broader guarantees, but disputes are typically resolved at the service-level SLA because that's where accountability directly lies with the provider's infrastructure.

QUESTION 4:

Which pair of metrics is most directly used to quantify the reliability and recoverability commitments in a cloud SLA?

- A. Availability (uptime %) and Mean Time To Recovery (MTTR)
- B. Lines of code and number of microservices
- C. Number of employees and office locations
- D. CPU clock speed and RAM size of the servers

Answer: A

Detailed Solution : Availability (uptime percentage) captures reliability, and Mean Time To Recovery (MTTR) captures how quickly the service is restored after an incident. The other options are organizational or low-level implementation details, not SLA commitments.

QUESTION 5:

A cloud service guarantees 99.9% availability per 30-day month. What is the maximum allowable downtime in that month under the SLA? (Assume 30 days = 720 hours.)

- A. About 0.72 hours
- B. About 7.2 hours



- C. About 3.6 hours
D. About 72 minutes

Answer: A

Detailed Solution : Allowed downtime = $(1 - 0.999) \times 720 \text{ hours} = 0.001 \times 720 = 0.72 \text{ hours} \approx 43 \text{ minutes per month.}$

QUESTION 6:

Which feature primarily enables a parallel database to process large analytical queries faster than a traditional database?

- A. Exclusive use of SQL
B. Distributed query execution across multiple computing nodes
C. Elimination of indexing
D. Storing all data in a single server

Correct Answer: B

Detailed Solution: Parallel databases partition data and execute query operations across multiple nodes simultaneously, significantly improving analytical performance.

QUESTION 7:

You are building a fraud-detection pipeline that must flag suspicious transactions within a few hundred milliseconds as they arrive. The system must:

- Ingest about 500,000 transactions per second
 - Detect and alert within sub-second latency
 - Apply per-transaction rules and short sliding-window aggregations
 - Which architectural choice is LEAST suitable?
- A. Apache Kafka feeding Apache Flink for continuous stream processing
B. Amazon Kinesis with a real-time stream-processing application
C. A nightly batch job on a data warehouse that scans the day's transactions
D. Apache Pulsar with a stream processor for windowed aggregation

Answer: C

Detailed Solution : A nightly batch job introduces hours of latency and cannot meet a sub-second, real-time detection requirement. Kafka + Flink, Kinesis real-time applications, and Pulsar + stream processors are all designed for low-latency continuous stream processing and are suitable here.

QUESTION 8:



A company stores petabytes of satellite imagery that must be processed in parallel by Spark clusters.

Which storage system is most suitable?

- A. Local SSD attached to a single virtual machine
- B. Object storage such as Amazon S3 or Google Cloud Storage
- C. Relational database server
- D. Client-side browser cache

Correct Answer: B

Detailed Solution: Cloud object storage provides virtually unlimited scalability, durability, and seamless integration with distributed processing frameworks like Spark and Hadoop.

QUESTION 9:

A MapReduce job is designed to count word frequencies in 2 TB of text files. The mapper output is 1.5 TB (including overhead), and after the shuffle and sort phase, the data is 1.3 TB. During the reduce phase, a node runs out of disk space during the merge operation. Which of the following is the most likely cause and solution?

- A. The mapper logic is inefficient; rewrite the mapper function
- B. The combiner phase didn't run; ensure combiner is explicitly configured to compress intermediate data locally
- C. The shuffle and sort didn't deduplicate keys; increase the number of reducers
- D. The network bandwidth is saturated; add more nodes to the cluster

Correct Answer: B

Detailed Solution : The combiner is a key MapReduce optimization that runs locally on each mapper node after the map phase, pre-aggregating intermediate key-value pairs to reduce their volume before shuffling. In a word count job, for example, it can sum occurrences of each word on the mapper node, potentially shrinking 1.5 TB of mapper output to 400–500 GB. Without a combiner, all intermediate data is sent to the reducers, which can overwhelm disk space during the merge phase. By explicitly configuring a combiner, the amount of data transferred across the network and merged by reducers is greatly reduced, preventing disk exhaustion and improving overall job efficiency.

QUESTION 10:

An OpenStack compute node contains 640 physical vCPUs. If the CPU allocation ratio is 5:1, what is the maximum number of virtual CPUs that may be allocated?



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- A. 3000
- B. 3200
- C. 3400
- D. 3500

Correct Answer: B

Detailed Solution: Total allocatable vCPUs = Physical vCPUs $\times$ Allocation Ratio = $640 \times 5 = 3200$ vCPUs
